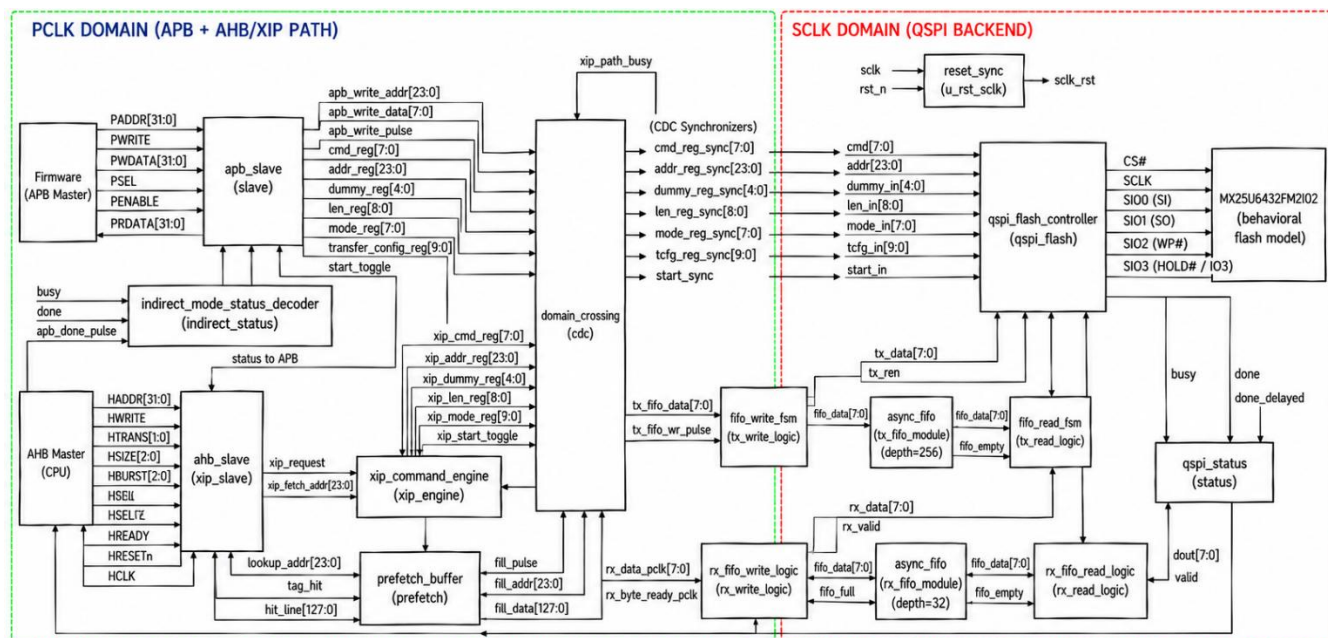


PROJECT – 5

QSPI Flash Controller with Execute-In-Place (XIP)

- G Gokul (EMP ID: 6892)

MICRO ARCHITECTURE



The entire project can be divided into two path, with each path trying to access the flash via the QSPI Flash Controller. The paths are:

1. **Indirect mode(APB):** Indirect mode is nothing but a generic command transfer engine. In indirect mode, Firmware feeds the input for APB slave. The firmware fills the register bank one by one and then starts the data transmission by enabling start flag in control status register present in the register bank. Once the transmission has started, the register data is sent to “domain_crossing” module, which then checks if the flash is busy and send performs enable signal based data crossing based on priority with “qspi_flash_controller” module via a toggle synchronizer since it is fast to slow domain crossing. The Flash Controller latches these values and sent them to the Behavioral Flash Model via the serial pins.
2. **XIP Mode:** In XIP mode, unlike Indirect mode, only address is given as input. The AHB Slave first checks if the prefetch buffer has the requested address data stored in it. If the data is present, then it is fetched and given back to AHB Master, else AHB slave has to access the Flash to read data.

REGISTER MAP

1. APB Register Bank

Offset (Hex)	Name	Reset value	Width	Access	Gating Condition	Description
0x0	command_code	0	8	RW	~indirect_path_busy	Opcode byte sent as the QSPI opcode phase
0x1	address_bytes	0	24	RW	~indirect_path_busy	Flash address for address-phase-enabled transactions. Future scope: Four byte address
0x2	dummy_cycles_count	0	5	RW	~indirect_path_busy	Dummy SCLK cycles count required for the particular flash command code
0x3	length_of_data	0	9	RW	~indirect_path_busy	Number of bytes to be written/read from the flash
0x4	control_signals	0	2	W	~indirect_path_busy & ~start_pending & ~xip_path_busy	LSB: Start transmission MSB: Abort transmission (future scope of work)
0x5	write_fifo_data	0	8	W	None — not gated by indirect_path_busy	Bytes of data to be written into the flash has to store into this register one by one.
0x6	read_fifo_data	0	8	R	Read of this offset clears rx_data_valid	During read operation, the flash stores output data in 'rx_fifo', and the byte from the 'rx_fifo' are loaded into this register one by one.
0x7	transfer_configuration_reg	0	10	RW	~indirect_path_busy	Phase-sequencer control bitfield; see transfer configuration bitfield table below. FSM is opcode-blind — entirely driven by this register
0x8	qspi_transfer_status_register	0	3	R	None — read-only status	{transmission_completed, indirect_path_busy, rx_data_valid} —

						packed into prdata[2:0], upper bits 0
0x9	mode_byte	0	8	RW	~indirect_path_bu sy	M7-M0 mode byte for flash transaction. Value depends on the behavioral flash module used.

2. Transfer Config Register of APB Register Bank

Bit Range	Field Name	Values	Description
[9:8]	data_phase_line_width	00 = 1-wire, 01 = 2-wire, 10 = 4-wire	I/O width during the data phase
[7:6]	addr_phase_line_width	00 = 1-wire, 01 = 2-wire, 10 = 4-wire	I/O width during the address/mode/dummy phases
[5]	addr_byte_count	0 = 3-byte address, 1 = 4-byte address	Selects 24-bit vs 32-bit address field width. Note: Current RTL only supports 3 byte address due to conflict with prefetch buffer
[4]	data_direction	0 = read from flash, 1 = write to flash	Direction of the data phase
[3]	addr_phase_enable	0 = skip, 1 = include	Whether an address phase follows the opcode
[2]	data_phase_enable	0 = skip, 1 = include	Whether a data phase is present
[1]	mode_phase_enable	0 = skip, 1 = include	Whether a mode-byte phase follows the address phase
[0]	dummy_phase_enable	0 = skip, 1 = include	Whether a dummy-cycle phase is present

RTL SOURCE

Directory structure:

qspi_flash_controller

- RTL
 - command_code.h
 - reset_sync.v
 - domain_crossing.v
 - async_fifo.v
 - fifo_write_fsm.v
 - fifo_read_fsm.v
 - rx_fifo_write_logic.v
 - rx_fifo_read_logic.v
 - qspi_status.v
 - qspi_flash_controller.v
 - prefetch_buffer.v
 - indirect_mode_status_decoder.v
 - apb_slave.v
 - ahb_slave.v
 - xip_command_engine.v
 - top_module.v

model

- MX25U6432FM2I02.v

tb

- tb.v

File list:

S.No	File	Description
1	command_code.h	Register offsets, command/opcode definitions, bit-field definitions and fixed XIP presets
2	reset_sync.v	Asynchronous reset assertion and synchronous reset deassertion for each clock domain
3	domain_crossing.v	Toggle-based PCLK-to-SCLK clock-domain crossing for the selected command/register set. Send commands to Flash Controller based on XIP and Indirect Mode request.
4	async_fifo.v	Dual-clock asynchronous FIFO using Gray-coded read/write pointers
5	fifo_write_fsm.v	Generates TX FIFO write control from APB FIFO writes
6	fifo_read_fsm.v	Reads TX FIFO data and supplies it to the QSPI Flash Controller
7	rx_fifo_write_logic.v	Captures valid QSPI RX data and generates RX FIFO write control
8	rx_fifo_read_logic.v	Reads received data from the RX FIFO and generates rx_byte_ready
9	qspi_status.v	Generates QSPI transaction busy, done and delayed-done status
10	qspi_flash_controller.v	Main QSPI serial transaction engine, drives serial pins of behavioral flash model
11	prefetch_buffer.v	One-line XIP prefetch/cache buffer. Each line contains 16 bytes of data. AHB Slave checks this prefetch buffer, if tag is present and is valid, then data is fetch from prefetch buffer, else data for the new address is fetched from the flash and stored back in prefetch buffer. If Indirect mode writes data into a address, and Prefetch Buffer happens to store data of the same address, then that data is invalidated.
12	indirect_mode_status_decoder.v	Generates indirect-path busy and transmission-completed status, which is then sent to PCLK domain via dual rank synchronizer. These signals are used to write into "qspi_transfer_status_register" of the APB Register Bank.
13	apb_slave.v	APB interface and indirect-mode register bank
14	ahb_slave.v	AHB-Lite interface and XIP request handling

15	xip_command_engine.v	Generates the command/configuration required for an XIP Flash read
16	top_module.v	Top-level integration of APB, AHB/XIP, ownership mux, CDC, FIFOs, QSPI controller and status logic

SELF CHECKING TESTBENCH



tb.v

Note: The failures reported by the self-checking testbench are due to minor errors in the testbench. The RTL functionality is correct, and no corresponding failures are observed in the simulation waveform.

Output Log:

VCD info: dumpfile wave.vcd opened for output.

[0] flash WIP=0 WEL=0 QE=0

[8005200] ---- RDSR1 (post-reset, expect WEL=0/WIP=0) ----

[8032300] PASS : RDSR1 post-reset WIP==0 (expected=0 actual=0)

[8032300] PASS : RDSR1 post-reset WEL==0 (expected=0 actual=0)

[8032300] ---- WREN ----

[8046800] flash WIP=0 WEL=1 QE=0

[8053100] ---- RDSR1 (post-WREN, expect WEL=1) ----

[8080300] PASS : RDSR1 post-WREN WEL==1 (expected=1 actual=1)

[8080300] ---- RDID ----

[8120300] PASS : RDID last byte (memory density) (expected=37 actual=37)

[8120300] ---- WREN (before page program @0x10) ----

[8141100] ---- PAGE PROGRAM (12 bytes @ 0x000010) ----

[8258800] flash WIP=1 WEL=1 QE=0

[12258800] flash WIP=0 WEL=0 QE=0

[13265100] flash[0x10] = ad

[13265100] flash[0x11] = be

[13265100] flash[0x12] = ef

[13265100] flash[0x13] = 42

[13265100] flash[0x14] = ad

[13265100] flash[0x15] = be
[13265100] flash[0x16] = ef
[13265100] flash[0x17] = 42
[13265100] flash[0x18] = ad
[13265100] flash[0x19] = be
[13265100] flash[0x1a] = ef
[13265100] flash[0x1b] = 11
[13265100] PASS : direct readback last byte @0x1B (expected=11 actual=11)
[13265100] ---- WREN (before FAST READ tests) ----
[13279600] flash WIP=0 WEL=1 QE=0
[13285900] ---- FAST READ (12 bytes @ 0x000010, 7 dummy) ----
[13409100] PASS : FAST_READ last byte @0x10 (expected=11 actual=11)
[13526800] flash WIP=1 WEL=1 QE=0
[17526800] flash WIP=0 WEL=0 QE=0
[18533100] ---- DUAL OUTPUT READ (DOR 0x3B, 12 bytes @ 0x000030, 7 dummy) ----
[18617900] PASS : DOR last byte @0x30 (expected=0b actual=0b)
[18617900] ---- DUAL I/O READ (DIO 0xBB, 12 bytes @ 0x000030, 3 dummy) ----
[18689900] PASS : DIO last byte @0x30 (expected=0b actual=0b)
[18689900] ---- WREN (before PP@0x70) ----
[18704400] flash WIP=0 WEL=1 QE=0
[18828400] flash WIP=1 WEL=1 QE=0
[22828400] flash WIP=0 WEL=0 QE=0
[23834700] ---- WREN (before WRSR) ----
[23849200] flash WIP=0 WEL=1 QE=0
[23879600] flash WIP=1 WEL=1 QE=0
[423879600] flash WIP=0 WEL=0 QE=1
[433885900] PASS : WRSR sets QE==1 (expected=1 actual=1)
[433885900] ---- RDSR1 (post-WRSR) ----

[433913100] PASS : R1 post-WRSR QE bit readback ==1 (expected=1 actual=1)

[433913100] ---- QUAD OUTPUT READ (QOR 0x6B, 12 bytes @ 0x000070, 7 dummy) ----

[433978700] PASS : QOR last byte @0x70 (expected=bb actual=bb)

[433978700] ---- QUAD I/O READ (QIOR 0xEB, 12 bytes @ 0x000070, 5 dummy) ----

[433978700] NOTE: mode phase is a no-op here -- top ties

 qspi_flash_controller.mode_reg to a constant 0 placeholder 43397870

 (no APB path for mode_byte yet). 43397870

[434028300] PASS : QIOR last byte @0x70 (expected=bb actual=bb)

[434028300] ---- WREN (before QIPP) ----

[434042800] flash WIP=0 WEL=1 QE=1

[434049100] ---- QUAD INPUT PAGE PROGRAM (QIPP 0x32, 4 bytes @ 0x000100) ----

[434077200] flash WIP=1 WEL=1 QE=1

[438077200] flash WIP=0 WEL=0 QE=1

[439083500] ---- WREN (before 256B page program @0x200) ----

[439098000] flash WIP=0 WEL=1 QE=1

[439104300] ---- PAGE PROGRAM (256 bytes @ 0x000200, incrementing pattern) ----

[440930000] flash WIP=1 WEL=1 QE=1

[444930000] flash WIP=0 WEL=0 QE=1

[445936300] PASS : PP 256B direct readback @0x200 (256 bytes verified, incrementing pattern)

[445936300] ---- WREN (before 256B QUAD OUTPUT READ) ----

[445950800] flash WIP=0 WEL=1 QE=1

[445957100] ---- QUAD OUTPUT READ (QOR 0x6B, 256 bytes @ 0x000200, 7 dummy) ----

[446413100] PASS : OR 256B streamed readback @0x200 (256 bytes verified, incrementing pattern)

[446413100] ===== XIP / AHB PATH TESTS =====

[446413100] ---- [XIP] Cold-miss BYTE read @0x000200 ----

[446485510] PASS : -miss byte read @0x200 HRESP==OK (expected=0 actual=0)

[446485510] PASS : XIP cold-miss byte read @0x200 (expected=00000000 actual=00000000)

[446485510] ---- [XIP] Cache-hit BYTE read @0x000205 ----

[446486510] PASS : e-hit byte read @0x205 HRESP==OK (expected=0 actual=0)

[446486510] PASS : XIP cache-hit byte read @0x205 (expected=00000005 actual=00000005)

[446486510] ---- [XIP] Cache-hit HALFWORD read @0x000208 ----

[446487510] PASS : t halfword read @0x208 HRESP==OK (expected=0 actual=0)

[446487510] PASS : P cache-hit halfword read @0x208 (expected=00000908 actual=00000908)

[446487510] ---- [XIP] Cache-hit WORD read @0x00020C ----

[446488510] PASS : e-hit word read @0x20C HRESP==OK (expected=0 actual=0)

[446488510] PASS : XIP cache-hit word read @0x20C (expected=0f0e0d0c actual=0f0e0d0c)

[446488510] ---- [XIP] Cold-miss WORD read @0x000230 (new line) ----

[446560710] PASS : -miss word read @0x230 HRESP==OK (expected=0 actual=0)

[446560710] PASS : XIP cold-miss word read @0x230 (expected=33323130 actual=33323130)

[446560710] ---- [XIP] Write-overlap invalidation @0x000230 ----

[446560710] PASS : lid before overlapping APB write (expected=1 actual=1)

[446560710] ---- WREN (before overlap PP @0x230) ----

[446726800] flash WIP=1 WEL=1 QE=1

[450726800] flash WIP=0 WEL=0 QE=1

[451733100] PASS : ed after overlapping APB program (expected=0 actual=0)

[451733100] ---- [XIP] Re-read @0x000230 after invalidation (must fetch updated data) ---
-

[451805510] PASS : alidate+refetch @0x230 HRESP==OK (expected=0 actual=0)

[451805510] PASS : after invalidate+refetch @0x230 (expected=23222120 actual=23222120)

[451805510] ---- [XIP] AHB WRITE attempt @0x000200 (expect HRESP=ERROR) ----

[451806310] PASS : IP region rejected (HRESP=ERROR) (expected=1 actual=1)

```

[451806310] PASS : pt left flash contents unchanged (expected=00 actual=00)
[451806310] ---- [XIP] Concurrent APB QOR (256B) + AHB read while indirect_path_busy=1
----
[452261900] FAIL : ill correct under XIP contention -- byte[48] expected=30 actual=20
[452261900] FAIL : ill correct under XIP contention -- byte[49] expected=31 actual=21
[452261900] FAIL : ill correct under XIP contention -- byte[50] expected=32 actual=22
[452261900] FAIL : ill correct under XIP contention -- byte[51] expected=33 actual=23
[452261900] FAIL : ill correct under XIP contention -- byte[52] expected=34 actual=24
[452261900] FAIL : ill correct under XIP contention -- byte[53] expected=35 actual=25
[452261900] FAIL : ill correct under XIP contention -- byte[54] expected=36 actual=26
[452261900] FAIL : ill correct under XIP contention -- byte[55] expected=37 actual=27
[452261900] FAIL : ill correct under XIP contention -- byte[56] expected=38 actual=28
[452261900] FAIL : ill correct under XIP contention -- byte[57] expected=39 actual=29
[452261900] FAIL : ill correct under XIP contention -- byte[58] expected=3a actual=2a
[452261900] FAIL : ill correct under XIP contention -- byte[59] expected=3b actual=2b
[452261900] FAIL : ill correct under XIP contention -- byte[60] expected=3c actual=2c
[452261900] FAIL : ill correct under XIP contention -- byte[61] expected=3d actual=2d
[452261900] FAIL : ill correct under XIP contention -- byte[62] expected=3e actual=2e
[452261900] FAIL : ill correct under XIP contention -- byte[63] expected=3f actual=2f
[452331910] PASS : irect_path_busy @0x280 HRESP==OK (expected=0 actual=0)
[452331910] PASS : behind indirect_path_busy @0x280 (expected=83828180
actual=83828180)
[452331910] ---- [XIP] Non-overlapping APB write must not invalidate cached line @0x280
----
[452331910] ---- WREN (before non-overlap PP @0x300) ----
[452346000] flash WIP=0 WEL=1 QE=1
[452414000] flash WIP=1 WEL=1 QE=1
[456414000] flash WIP=0 WEL=0 QE=1
[457420300] PASS : after non-overlapping APB write (expected=1 actual=1)

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[457421110] PASS : pping APB write @0x285 HRESP==OK (expected=0 actual=0)

[457421110] PASS : non-overlapping APB write @0x285 (expected=87868584 actual=87868584)

[457421110] ===== AHB NONSEQ -> SEQ TESTS =====

[457421110] ---- [XIP] Legal NONSEQ -> SEQ 4-beat burst @0x200 ----

[457421320] PASS : SEQ burst @0x200 beat0 HRESP==OK (expected=0 actual=0)

[457421320] FAIL : ONSEQ->SEQ burst @0x200 beat0 @0 (expected=03020100 actual=87868584)

[457492910] PASS : SEQ burst @0x200 beat1 HRESP==OK (expected=0 actual=0)

[457492910] PASS : ONSEQ->SEQ burst @0x200 beat1 @4 (expected=07060504 actual=07060504)

[457493120] PASS : SEQ burst @0x200 beat2 HRESP==OK (expected=0 actual=0)

[457493120] FAIL : ONSEQ->SEQ burst @0x200 beat2 @8 (expected=0b0a0908 actual=07060504)

[457493710] PASS : SEQ burst @0x200 beat3 HRESP==OK (expected=0 actual=0)

[457493710] PASS : ONSEQ->SEQ burst @0x200 beat3 @C (expected=0f0e0d0c actual=0f0e0d0c)

[457494110] ---- [XIP] Legal NONSEQ -> SEQ burst crossing 0x20F/0x210 ----

[457494320] PASS : ary burst @0x20C beat0 HRESP==OK (expected=0 actual=0)

[457494320] PASS : Q boundary burst @0x20C beat0 @0 (expected=0f0e0d0c actual=0f0e0d0c)

[457494910] PASS : ary burst @0x20C beat1 HRESP==OK (expected=0 actual=0)

[457494910] FAIL : Q boundary burst @0x20C beat1 @4 (expected=13121110 actual=03020100)

[457566510] PASS : ary burst @0x20C beat2 HRESP==OK (expected=0 actual=0)

[457566510] PASS : Q boundary burst @0x20C beat2 @8 (expected=17161514 actual=17161514)

[457566720] PASS : ary burst @0x20C beat3 HRESP==OK (expected=0 actual=0)

[457566720] FAIL : Q boundary burst @0x20C beat3 @C (expected=1b1a1918 actual=17161514)

[457567110] ---- [XIP] New NONSEQ burst after IDLE ----

[457639110] PASS : after SEQ burst @0x220 HRESP==OK (expected=0 actual=0)

[457639110] PASS : ew NONSEQ after SEQ burst @0x220 (expected=23222120 actual=23222120)

[457639110] ===== XIP CORNER-CASES =====

[457639110] ---- [XIP] Byte read at line boundary offset 15 @0x20F ----

[457711110] PASS : st byte of line 0x200) HRESP==OK (expected=0 actual=0)

[457711110] PASS : @0x20F (last byte of line 0x200) (expected=0000000f actual=0000000f)

[457711110] ---- [XIP] Byte read at next line start @0x210 ----

[457783110] PASS : st byte of line 0x210) HRESP==OK (expected=0 actual=0)

[457783110] PASS : 0x210 (first byte of line 0x210) (expected=00000010 actual=00000010)

[457783110] ---- [XIP] Halfword read @0x210 (aligned) ----

[457784110] PASS : P halfword read @0x210 HRESP==OK (expected=0 actual=0)

[457784110] PASS : XIP halfword read @0x210 (expected=00001110 actual=00001110)

[457784110] ---- [XIP] Word read @0x210 (aligned) ----

[457785110] PASS : XIP word read @0x210 HRESP==OK (expected=0 actual=0)

[457785110] PASS : XIP word read @0x210 (expected=13121110 actual=13121110)

[457785110] ---- [XIP] Byte read @0x215 (inside line 0x210) ----

[457786110] PASS : XIP byte read @0x215 HRESP==OK (expected=0 actual=0)

[457786110] PASS : XIP byte read @0x215 (expected=00000015 actual=00000015)

[457786110] ---- [XIP] Word read @0x21C (offset 12, aligned) ----

[457787110] PASS : XIP word read @0x21C HRESP==OK (expected=0 actual=0)

[457787110] PASS : XIP word read @0x21C (expected=1f1e1d1c actual=1f1e1d1c)

[457787110] ---- [XIP] Cold-miss @0x000300 (new line) ----

[457859110] PASS : XIP cold-miss @0x300 HRESP==OK (expected=0 actual=0)

[457859110] PASS : XIP cold-miss @0x300 (expected=5d5c5b5a actual=5d5c5b5a)

[457859110] ---- [XIP] Cold-miss @0x000400 (new line, evicts 0x300) ----

[457931110] PASS : XIP cold-miss @0x400 HRESP==OK (expected=0 actual=0)

[457931110] PASS : XIP cold-miss @0x400 (expected=ffffffff actual=ffffffff)

[457931110] ---- [XIP] Re-read @0x000300 (must refetch, was evicted) ----

[458003110] PASS : @0x300 after eviction HRESP==OK (expected=0 actual=0)

[458003110] PASS : IP re-read @0x300 after eviction (expected=5d5c5b5a actual=5d5c5b5a)

[458003110] ---- [XIP] Long erase + XIP read (polling test) ----

[458003110] ---- [XIP] Prime line @0x500 (cold miss) ----

[458075110] PASS : XIP cold-miss @0x500 HRESP==OK (expected=0 actual=0)

[458075110] PASS : XIP cold-miss @0x500 (expected=ffffff actual=ffffff)

[458075110] ---- WREN (before long SECTOR ERASE @0x1000) ----

[458089200] flash WIP=0 WEL=1 QE=1

[458095500] ---- Start sector erase @0x1000 (WIP will be high) ----

[458100300] ---- [XIP] Read @0x600 during erase (should poll WIP) ----

[458129200] flash WIP=1 WEL=1 QE=1

[758129200] flash WIP=0 WEL=0 QE=1

[758203910] PASS : ng erase (WIP polling) HRESP==OK (expected=0 actual=0)

[758203910] PASS : 0x600 during erase (WIP polling) (expected=ffffff actual=ffffff)

[758203910] ---- [XIP] Post-erase, verify that the line @0x600 was fetched correctly ----

[758204910] PASS : after erase completed HRESP==OK (expected=0 actual=0)

[758204910] PASS : ead @0x600 after erase completed (expected=ffffff actual=ffffff)

[758204910] ===== END XIP CORNER-CASES =====

[758204910] ---- WREN (before SECTOR ERASE @0x700) ----

[758219600] flash WIP=0 WEL=1 QE=1

[758225900] ---- SECTOR ERASE (SE 0x20) @ 0x000700 ----

[758259600] flash WIP=1 WEL=1 QE=1

[1058259600] flash WIP=0 WEL=0 QE=1

[1058265900] ---- QUAD OUTPUT READ post-SE (QOR 0x6B, 12 bytes @ 0x000070, 7 dummy) ----

[1058331500] PASS : E erased region reads 0xFF @0x70 (expected=ff actual=ff)

[1058331500] ---- WREN (before BLOCK ERASE) ----

[1058346000] flash WIP=0 WEL=1 QE=1

[1058352300] ---- BLOCK ERASE (BE 0xD8) @ 0x000000 ----

[1058386000] flash WIP=1 WEL=1 QE=1

[4058386000] flash WIP=0 WEL=0 QE=1

[4158392300] ---- QUAD OUTPUT READ post-BE (QOR 0x6B, 12 bytes @ 0x000030, 7 dummy) ----

[4158457900] PASS : E erased region reads 0xFF @0x30 (expected=ff actual=ff)

[4158457900] ---- RDSR2 (0x35) ----

[4158457900] NOTE: this opcode is cmdEQIO on the flash model, not RDSR2 --

issuing it switches the model into QPI mode. 415845790

[4158478700] PASS : QIO switches flash into QPI mode (expected=1 actual=1)

=====

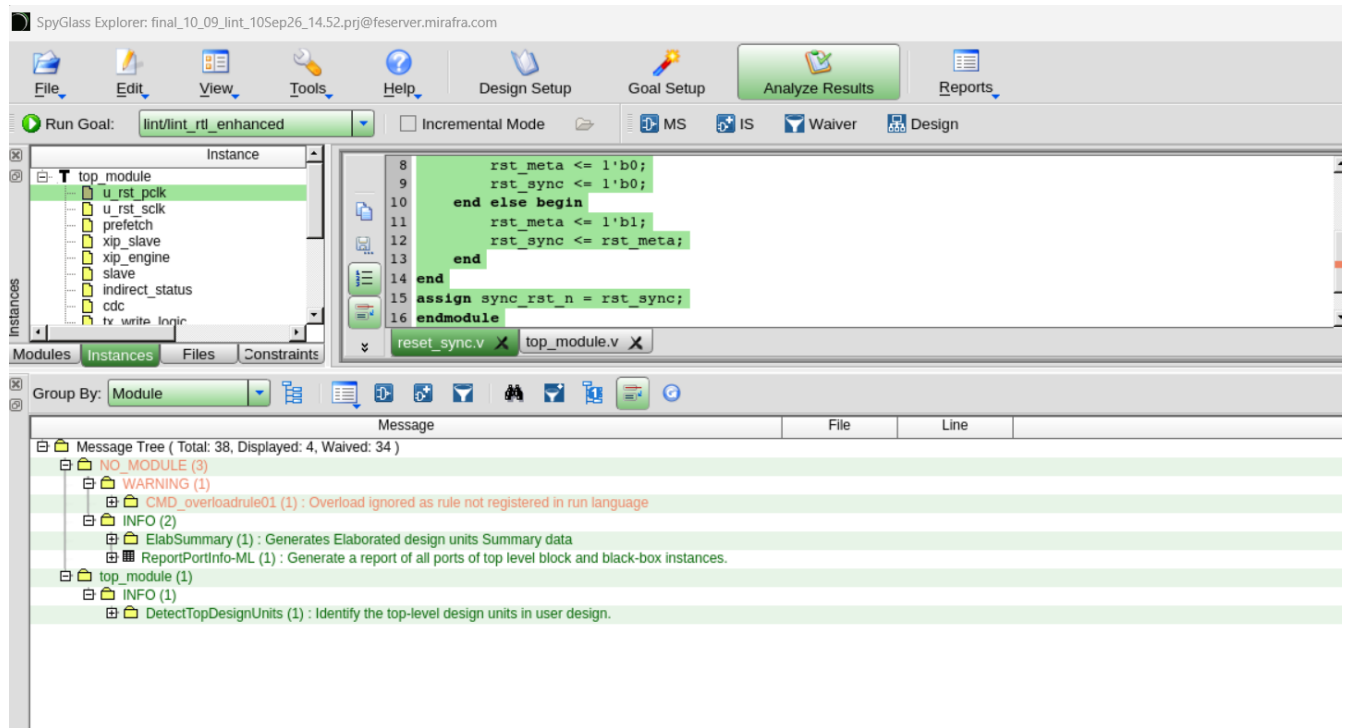
[4158478700] TESTBENCH RESULT: 5/81 CHECKS FAILED

=====

[4158478700] ---- Test sequence complete ----

tb.v:930: \$finish called at 4158488700 (100ps)

LINT REPORT



Lint RTL Enhanced Report:



moresimple.rpt

Waiver file:

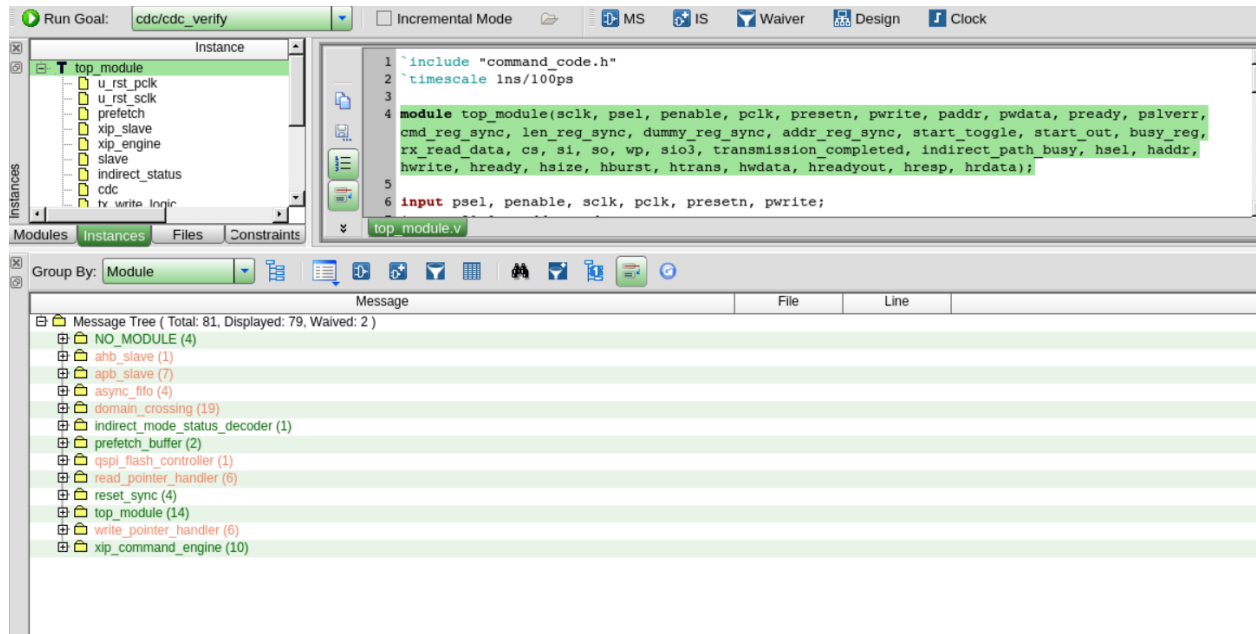


waiver.rpt

Justification to waive the corresponding errors:

1. **STARC05-2.5.1.2: Tristate buffer 'si' has logic in enable condition:** As per the Flash Behavioral Model Specification, it is mandatory to drive the serial pins of the flash to high impedance state.
2. **Signal io_in is assigned Z value**
3. **Tristate port 'si' detected**

CDC REPORT



Section E: Results Summary (Current)

RuleName	Passed	Failed	Partially Proved (Average Depth)	Not Analyzed	Others	Total
Ac_conv02	2	0	1(200)	0	0	3
Ac_conv02	0	0	2(200)	0	0	2
Ac_cdc01a	3	0	1(200)	0	0	4
Ac_datahold01a	1	0	7(200)	0	0	8
Total	6	0	11	0	0	17

All the warnings are partially proved which can be waived off. The reason being, as per the project specification, it is mandatory to use a handshake and FIFO in indirect mode. In FIFO, dual rank synchronizers are used on points, which then decides the FIFO full and empty conditions. Another enable based synchronizer is used via toggle synchronizer for PCLK to SCLK domain command transfer(cmd_reg, len_reg, etc). These two signals then converge in the “qspi_flash_controller” module. Hence, the tool is throwing a CDC warning for that case. Hence it has to be waived off.

CDC INVENTORY

S.No	Signal(s)	Direction	Synchronizing Structure	Location
1	start (indirect-mode kickoff)	PCLK → SCLK	Toggle flop (start_toggle) + 3-FF synchronizer chain + XOR edge detector	domain_crossing.v
2	xip_start_toggle (XIP-owned start, muxed with #1 before crossing)	PCLK → SCLK	Same toggle + 3FF + XOR structure in domain_crossing.v (mux happens before the crossing, in top_cdc_fifo_integrated.v's ownership mux)	domain_crossing.v
3	qspi_done (transaction-complete pulse)	SCLK → PCLK	Plain 2-FF synchronizer + edge detect	indirect_mode_status_decoder.v
4	qspi_busy	SCLK → PCLK	Plain 2-FF synchronizer	apb_slave.v (sync1 / qspi_busy_sync)
5	TX FIFO write pointer (firmware preload data)	PCLK → SCLK	Gray-coded pointer, dual-rank synchronized (async_fifo.v, write_pointer_handler / read_pointer_handler)	async_fifo #(.ADDR_WIDTH(8)) instance (tx_fifo_module)
6	RX FIFO write pointer (captured Flash read data)	SCLK → PCLK	Gray-coded pointer, dual-rank synchronized	async_fifo #(.ADDR_WIDTH(5)) instance (rx_fifo_module)
7	presetn (async reset)	Async → PCLK	2-FF async-assert/sync-deassert reset synchronizer	reset_sync.v (u_rst_pclk)
8	presetn (async reset)	Async → SCLK	2-FF async-assert/sync-deassert reset synchronizer	reset_sync.v (u_rst_sclk)

SDC CONSTRAINTS

```
create_clock -name pclk -period 20.0 -waveform {0.0 10.0} [get_ports pclk]
```

```
create_clock -name sclk -period 80.0 -waveform {0.0 40.0} [get_ports sclk]
```

```
set_clock_groups -asynchronous \
```

```
    -group {pclk} \
```

```
    -group {sclk}
```

```
set_false_path -from [get_ports presetn]
```

```
set_clock_uncertainty -setup 0.30 [get_clocks pclk]
```

```
set_clock_uncertainty -hold 0.15 [get_clocks pclk]
```

```
set_clock_uncertainty -setup 0.50 [get_clocks sclk]
```

```
set_clock_uncertainty -hold 0.2 [get_clocks sclk]
```

```
set_input_delay -clock pclk -max 2.0 [get_ports {psel penable pwrite}]
```

```
set_input_delay -clock pclk -min 0.5 [get_ports {psel penable pwrite}]
```

```
set_input_delay -clock pclk -max 2.0 [get_ports {paddr[*] pwrite[*]}]
```

```
set_input_delay -clock pclk -min 0.5 [get_ports {paddr[*] pwrite[*]}]
```

```
set_output_delay -clock pclk -max 2.0 [get_ports {pready pslverr}]
```

```
set_output_delay -clock pclk -min 0.5 [get_ports {pready pslverr}]
```

```
set_output_delay -clock pclk -max 2.0 [get_ports {prdata[*]}]
```

```
set_output_delay -clock pclk -min 0.5 [get_ports {prdata[*]}]
```

```
set_input_delay -clock pclk -max 2.0 [get_ports {hsel hwrite hready}]
```

```
set_input_delay -clock pclk -min 0.5 [get_ports {hsel hwrite hready}]
```

set_input_delay -clock pclk -max 2.0 [get_ports {hsize[*] hburst[*] htrans[*]}]

set_input_delay -clock pclk -min 0.5 [get_ports {hsize[*] hburst[*] htrans[*]}]

set_input_delay -clock pclk -max 2.0 [get_ports {haddr[*] hwddata[*]}]

set_input_delay -clock pclk -min 0.5 [get_ports {haddr[*] hwddata[*]}]

set_output_delay -clock pclk -max 2.0 [get_ports {hreadyout hresp}]

set_output_delay -clock pclk -min 0.5 [get_ports {hreadyout hresp}]

set_output_delay -clock pclk -max 2.0 [get_ports {hrdata[*]}]

set_output_delay -clock pclk -min 0.5 [get_ports {hrdata[*]}]

set_output_delay -clock sclk -max 5.0 [get_ports cs]

set_output_delay -clock sclk -min 1.0 [get_ports cs]

set_output_delay -clock sclk -max 5.0 [get_ports {si so wp sio3}]

set_output_delay -clock sclk -min 1.0 [get_ports {si so wp sio3}]

set_input_delay -clock sclk -max 6.0 [get_ports {si so wp sio3}]

set_input_delay -clock sclk -min 1.0 [get_ports {si so wp sio3}]

SYNTHESIS REPORTS

Report QOR

```
report_qor report_qtm_model
dc_shell> report_qor
*****
Report : qor
Design : top_module
Version: 5-2021.06-SP4
Date   : Thu Sep 10 15:23:38 2025
*****

Timing Path Group 'pclk'
-----
Levels of Logic:      6.00
Critical Path Length:  0.84
Critical Path Slack:   15.77
Critical Path Clk Period: 20.00
Total Negative Slack:  0.00
No. of Violating Paths: 0.00
Worst Hold Violation:  0.00
Total Hold Violation:  0.00
No. of Hold Violations: 0.00
-----

Timing Path Group 'sclk'
-----
Levels of Logic:      12.00
Critical Path Length:  1.39
Critical Path Slack:   73.11
Critical Path Clk Period: 80.00
Total Negative Slack:  0.00
No. of Violating Paths: 0.00
Worst Hold Violation:  0.00
Total Hold Violation:  0.00
No. of Hold Violations: 0.00
-----

Cell Count
-----
Hierarchical Cell Count: 2
Hierarchical Port Count: 113
Leaf Cell Count: 9595
Buf/Inv Cell Count: 584
Buf Cell Count: 72
Inv Cell Count: 512
CT Buf/Inv Cell Count: 0
Combinational Cell Count: 6464
Sequential Cell Count: 3131
Macro Count: 0
-----

Area
-----
Combinational Area: 15689.524516
Noncombinational Area: 22294.828255
Buf/Inv Area: 909.835522
Total Buffer Area: 241.44
Total Inverter Area: 669.40
Macro/Block Box Area: 0.000000
Net Area: 19671.551253
-----
Cell Area: 37903.544871
Design Area: 57575.096124
-----

Design Rules
-----
Total Number of Nets: 5836
Nets With Violations: 11
Max Trans Violations: 0
Max Cap Violations: 11
-----

Hostname: feserver.mirafr.com

Compile CPU Statistics
-----
Resource Sharing: 2.87
Logic Optimization: 1.27
Mapping Optimization: 9.80
-----
Overall Compile Time: 42.51
Overall Compile Wall Clock Time: 44.01
-----

Design WNS: 0.00 TNS: 0.00 Number of Violating Paths: 0

Design (Hold) WNS: 0.00 TNS: 0.00 Number of Violating Paths: 0
```

Check Design

```
check_design

*****
check_design summary:
Version:    S-2021.06-SP4
Date:      Fri Sep 11 12:58:17 2026
*****

-----
Name                                     Total
-----
Inputs/Outputs                          83
  Unconnected ports (LINT-28)           80
  Shorted outputs (LINT-31)              1
  Constant outputs (LINT-52)             2
Tristate                                4
  Single tristate driver drives an input pin (LINT-63) 4
-----
```


Report timing

```
*****
Report : timing
-path full
-delay max
-max_paths 1
Design : top_module
Version : S-2021.06-SP4
Date : Fri Sep 11 12:56:54 2025
*****

# A fanout number of 1000 was used for high fanout net computations.

Operating Conditions: ss8p75v25c Library: saed32lvt_ss8p75v25c
Wire Load Model Mode: enclosed

Startpoint: paddr[3] (input port clocked by pclk)
Endpoint: slave/addr_reg_reg[0]
(rising edge-triggered flip-flop clocked by pclk)
Path Group: pclk
Path Type: max

Des/Clust/Port Wire Load Model Library
-----
top_module 70000 saed32lvt_ss8p75v25c

Point Incr Path
-----
clock pclk (rise edge) 0.00 0.00
clock network delay (ideal) 0.00 0.00
input external delay 2.00 2.00 f
paddr[3] (in) 0.00 2.00 f
U1768/Y (INVX8_LVT) 0.02 2.02 r
U2148/Y (AND4X1_LVT) 0.15 2.17 r
U2181/Y (NAND4X8_LVT) 0.21 2.38 f
U1731/Y (INVX2_LVT) 0.21 2.59 r
U1720/Y (INVX8_LVT) 0.13 2.72 f
U2182/Y (AO22X1_LVT) 0.12 2.84 f
slave/addr_reg_reg[0]/D (OFFARX1_LVT) 0.00 2.84 f
data arrival time 2.84

clock pclk (rise edge) 20.00 20.00
clock network delay (ideal) 0.00 20.00
clock uncertainty -0.30 19.70
slave/addr_reg_reg[0]/CLK (OFFARX1_LVT) 0.00 19.70 r
library setup time -0.10 19.60
data required time 19.60
-----
data required time 19.60
data arrival time -2.84
-----
slack (MET) 16.77

Startpoint: qspi_flash/count_bits_sent_reg[17]
(rising edge-triggered flip-flop clocked by sclk)
Endpoint: so (output port clocked by sclk)
Path Group: sclk
Path Type: max

Des/Clust/Port Wire Load Model Library
-----
top_module 70000 saed32lvt_ss8p75v25c
qspi_flash_controller 8000 saed32lvt_ss8p75v25c

Point Incr Path
-----
clock sclk (rise edge) 0.00 0.00
clock network delay (ideal) 0.00 0.00
qspi_flash/count_bits_sent_reg[17]/CLK (OFFARX1_LVT) 0.00 0.00 r
qspi_flash/count_bits_sent_reg[17]/Q1 (OFFARX1_LVT) 0.16 0.16 r
qspi_flash/U40/Y (AND4X1_LVT) 0.11 0.27 r
qspi_flash/U41/Y (NAND3X8_LVT) 0.05 0.32 f
qspi_flash/U42/Y (NOR4X1_LVT) 0.16 0.48 r
qspi_flash/U43/Y (NAND2X8_LVT) 0.05 0.52 f
qspi_flash/U45/Y (NOR4X1_LVT) 0.19 0.72 r
qspi_flash/U47/Y (NAND4X8_LVT) 0.06 0.78 f
qspi_flash/U48/Y (OR3X1_LVT) 0.17 0.95 f
qspi_flash/U50/Y (AO22X1_LVT) 0.11 1.07 f
qspi_flash/U55/Y (NAND2X8_LVT) 0.11 1.17 r
qspi_flash/U77/Y (INVX8_LVT) 0.03 1.20 f
qspi_flash/U83/Y (NAND3X8_LVT) 0.07 1.27 r
qspi_flash/so_tru/Y (TNBUFX1_LVT) 0.12 1.39 f
qspi_flash/so (qspi_flash_controller) 0.00 1.39 f
so (trout) 0.00 1.39 f
data arrival time 1.39

clock sclk (rise edge) 80.00 80.00
clock network delay (ideal) 0.00 80.00
clock uncertainty -0.50 79.50
output external delay -5.00 74.50
data required time 74.50
-----
data required time 74.50
data arrival time -1.39
-----
slack (MET) 73.11
```

Report Area

```
1
dc_shell> report_area

*****
Report : area
Design : top_module
Version: S-2021.06-SP4
Date   : Fri Sep 11 12:56:19 2026
*****

Information: Updating design information... (UID-85)
Warning: Design 'top_module' contains 1 high-fanout nets. A fanout number of 1000 will be used for delay calculations involving these nets. (TIM-134)
Library(s) Used:

    saed32lvt_ss0p75v25c (File: /fetools/work_area/frontend/SAED32_EDK/lib/stdcell_lvt/db_nldm/saed32lvt_ss0p75v25c.db)

Number of ports:          358
Number of nets:          9948
Number of cells:         9599
Number of combinational cells: 6460
Number of sequential cells: 3135
Number of macros/black boxes: 0
Number of buf/inv:       584
Number of references:     40

Combinational area:      15609.524516
Buf/Inv area:            909.835522
Noncombinational area:   22294.020355
Macro/Black Box area:    0.000000
Net Interconnect area:   19671.551253

Total cell area:         37903.544871
Total area:              57575.096124
1
dc_shell> █
```

Netlist



netlist_qspi.v

STA REPORTS

SETUP TIMING

```
pt_shell> report_timing -delay_type max
*****
Report : timing
        -path_type full
        -delay_type max
        -max_paths 1
        -sort_by slack
Design : top_module
Version: S-2021.06-SP2
Date   : Fri Sep 11 14:09:09 2026
*****

Startpoint: paddr[3] (input port clocked by pclk)
Endpoint:  slave/addr_reg_reg[0]
          (rising edge-triggered flip-flop clocked by pclk)
Path Group: pclk
Path Type: max

Point                                     Incr      Path
-----
clock pclk (rise edge)                   0.00       0.00
clock network delay (ideal)              0.00       0.00
input external delay                     2.00       2.00 f
paddr[3] (in)                             0.00       2.00 f
U1768/Y (INVX0_LVT)                      0.02       2.02 r
U2148/Y (AND4X1_LVT)                     0.15       2.17 r
U2181/Y (NAND4X0_LVT)                    0.21       2.38 f
U1731/Y (INVX2_LVT)                      0.21       2.59 r
U1720/Y (INVX0_LVT)                      0.13       2.72 f
U2182/Y (A022X1_LVT)                     0.12       2.84 f
slave/addr_reg_reg[0]/D (DFFARX1_LVT)    0.00       2.84 f
data arrival time                        2.84

clock pclk (rise edge)                   20.00      20.00
clock network delay (ideal)              0.00      20.00
clock reconvergence pessimism             0.00      20.00
clock uncertainty                         -0.30      19.70
slave/addr_reg_reg[0]/CLK (DFFARX1_LVT)  19.70 r
library setup time                       -0.10      19.60
data required time                       19.60
-----
data required time                       19.60
data arrival time                        -2.84
-----
slack (MET)                             16.77

1
```

HOLD CHECK

Initially there was a hold violation of -0.16, then I inserted buffers in the path manually to increase data arrival time, as a result hold slack increased to 0.2.

```
Startpoint: u_rst_sclk/rst_sync_reg
(rising edge-triggered flip-flop clocked by sclk)
Endpoint: cdc/cmd_reg_sync_reg[4]
(remove check against rising-edge clock sclk)
Path Group: **async_default**
Path Type: min

Point                                Incr                                Path
-----
clock sclk (rise edge)                0.00                                0.00
clock network delay (ideal)            0.00                                0.00
u_rst_sclk/rst_sync_reg/CLK (DFFARX1_LVT) 0.00                                0.00 r
u_rst_sclk/rst_sync_reg/Q (DFFARX1_LVT) 0.22                                0.22 r
U16/Y (NBUFFX16_LVT)                  0.07                                0.29 r
U9/Y (NBUFFX4_LVT)                    0.09                                0.38 r
cdc/cmd_reg_sync_reg[4]/RSTB (DFFARX1_LVT) 0.00                                0.38 r
data arrival time                      0.38
-----
clock sclk (rise edge)                0.00                                0.00
clock network delay (ideal)            0.00                                0.00
clock reconvergence pessimism          0.00                                0.00
clock uncertainty                      0.20                                0.20
cdc/cmd_reg_sync_reg[4]/CLK (DFFARX1_LVT) 0.00                                0.20 r
library removal time                  0.19                                0.39
data required time                    0.39
-----
data required time                    0.39
data arrival time                      -0.38
-----
slack (VIOLATED)                      -0.01

Warning: report_timing has satisfied the max_paths criteria. There are 5 further endpoints which were ignored when generating this report. (UITE-502)

1
ot_shell> insert_buffer [get_pins u_rst_sclk/rst_sync_reg/Q] NBUFFX16_LVT
Information: Invalidating logical update. (PTE-139)
Warning: 'U21' not found in any path collections
Information: Inserted 'U21' at 'u_rst_sclk/rst_sync_reg/Q' with 'saed32lvt_ss0p75v25c/NBUFFX16_LVT'
{"U21"}
ot_shell> update_timing
Warning: Undo cleared by 'update_timing' (GUI-998)
1
ot_shell> report_timing -delay_type min -nworst 2
*****
Report : timing
-path_type full
-delay_type min
-nworst 2
-slack_lesser_than 0.00
-max_paths 2
-sort_by slack
Design : top_module
Version: S-2021.06-SP2
Date   : Fri Sep 11 14:55:12 2026
*****
No paths with slack less than 0.00.
```

In the above image you can see there was a slack of -0.01 even after adding buffer, but another buffer was inserted NBUFFX16_LVT, then there was no path with slack less than 0.

```

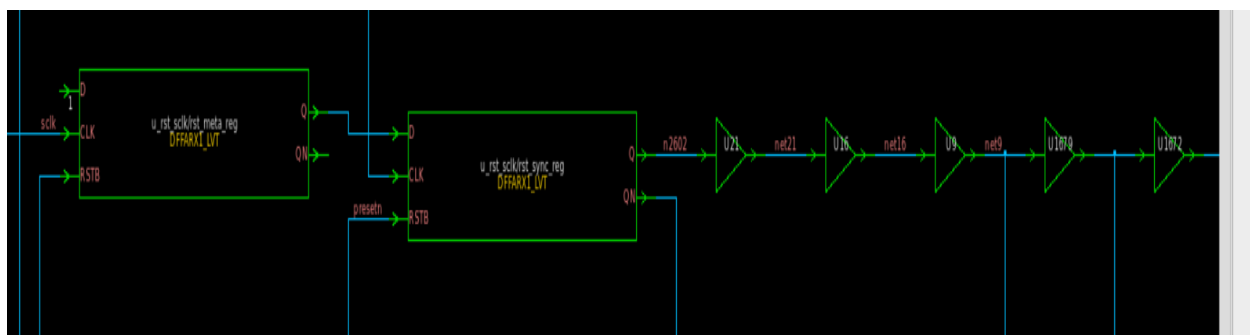
pt_shell> report_timing -delay_type min
*****
Report : timing
        -path_type full
        -delay_type min
        -max_paths 1
        -sort_by slack
Design : top_module
Version: S-2021.06-SP2
Date   : Fri Sep 11 14:57:01 2026
*****

Startpoint: cdc/apb_sync1_reg
            (rising edge-triggered flip-flop clocked by sclk)
Endpoint:  cdc/apb_sync2_reg
            (rising edge-triggered flip-flop clocked by sclk)
Path Group: sclk
Path Type: min

Point                                           Incr      Path
-----
clock sclk (rise edge)                        0.00      0.00
clock network delay (ideal)                   0.00      0.00
cdc/apb_sync1_reg/CLK (DFFARX1_LVT)           0.00      0.00 r
cdc/apb_sync1_reg/Q (DFFARX1_LVT)            0.19      0.19 f
cdc/apb_sync2_reg/D (DFFARX1_LVT)            0.00      0.19 f
data arrival time                             0.19
-----
clock sclk (rise edge)                        0.00      0.00
clock network delay (ideal)                   0.00      0.00
clock reconvergence pessimism                 0.00      0.00
clock uncertainty                             0.20      0.20
cdc/apb_sync2_reg/CLK (DFFARX1_LVT)           0.20      0.20 r
library hold time                           -0.03      0.17
data required time                            0.17
-----
data required time                            0.17
data arrival time                            -0.19
-----
slack (MET)                                   0.02
-----
1
pt_shell> 

```

Insertion of buffer is shown in GUI



REPORT POWER ESTIMATE

```
pt_shell> report_power
Warning: Neither event file or switching activity data present for power estimation. The command will propagate switching activity values for power calculation. (PWR-246)
Information: Running switching activity propagation with 4 threads!
Information: Running averaged power analysis... (PWR-601)
Information: Running power calculation with 4 threads. (PWR-602)
*****
Report : Averaged Power
Design : top_module
Version: S-2021.06-SP2
Date   : Fri Sep 11 15:01:40 2026
*****

Attributes
-----
t - Including register clock pin internal power
u - User defined power group

Power Group      Internal Power      Switching Power      Leakage Power      Total Power      (    %)      Attrs
-----
clock_network    4.711e-04      0.0000      0.0000  4.711e-04  (38.87%)  t
register         5.209e-07  4.914e-08  5.967e-04  5.973e-04  (49.28%)
combnational     1.140e-06  4.315e-06  1.381e-04  1.436e-04  (11.85%)
sequential       0.0000      0.0000      0.0000      0.0000      ( 0.00%)
memory           0.0000      0.0000      0.0000      0.0000      ( 0.00%)
io_pad           0.0000      0.0000      0.0000      0.0000      ( 0.00%)
black_box        0.0000      0.0000      0.0000      0.0000      ( 0.00%)

Net Switching Power = 4.364e-06 ( 0.36%)
Cell Internal Power = 4.727e-04 (39.01%)
Cell Leakage Power  = 7.349e-04 (60.63%)
-----
Total Power         = 1.212e-03 (100.00%)

1
pt_shell> []
```

REPORT WORST PATH OF HOLD

```
pt_shell> report_timing -delay_type min -nworst 2
*****
Report : timing
-path_type full
-delay_type min
-nworst 2
-slack_lesser_than 0.00
-max_paths 2
-sort_by slack
Design : top_module
Version: S-2021.06-SP2
Date   : Fri Sep 11 14:55:12 2026
*****

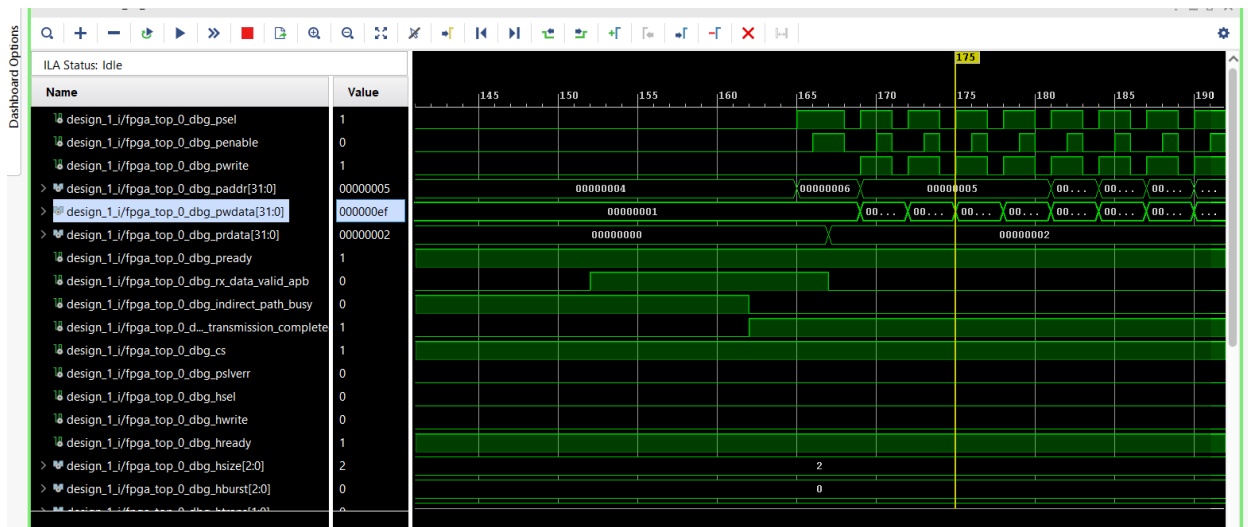
No paths with slack less than 0.00.
```

FPGA PROTOTYPE

ILA OUTPUT

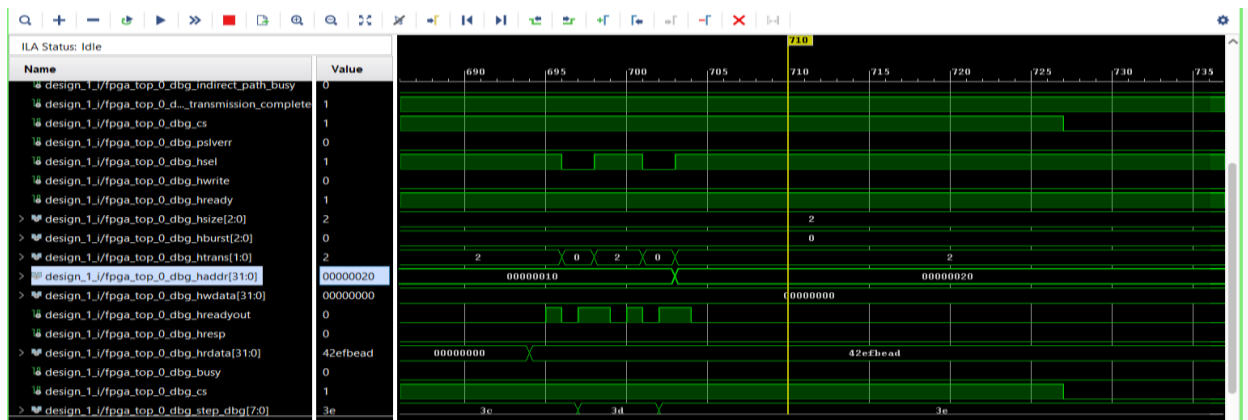
1. Command 0x05 sent to APB

Output of 0x02 is sent back via prdata[31:0]

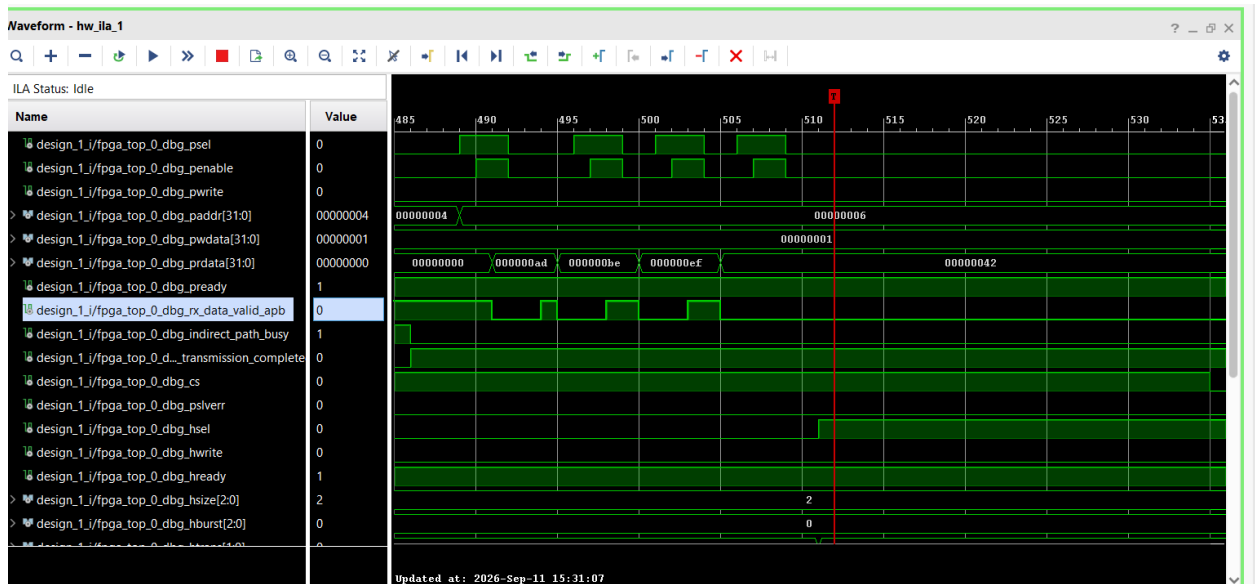


2. XIP MODE

Data is fetched from prefetch buffer and is sent back via hrdata[31:0]



3. READ FOUR BYTES OF DATA – OUTPUT RECEIVED BACK VIA prdata[31:0]



4. HTRANS IS SET TO NONSEQ

